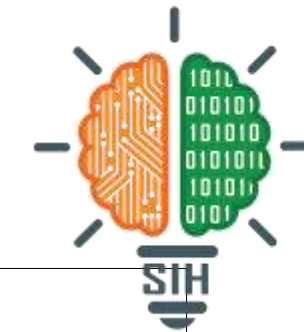


SMART INDIA HACKATHON 2026



**SMART INDIA
HACKATHON
2026**

TITLE PAGE

Problem Statement ID – SIH26206

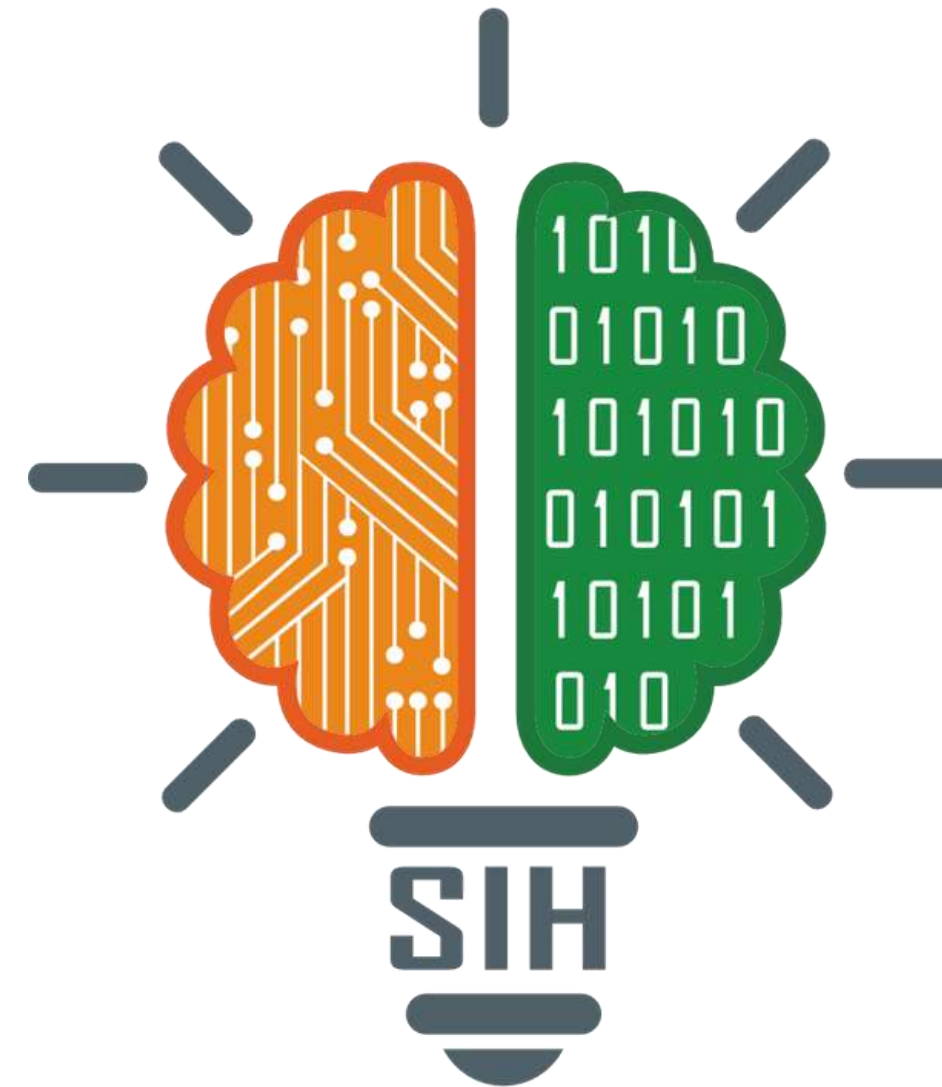
**Problem Statement Title – AI-Based Urban Waterlogging
Risk Prediction and Preventive Response System**

Theme – Disaster Management

PS Category – Software

Team ID – 157783

Team Name – INNOVOEX-1





THE PROBLEM

Reactive Waterlogging Response



Waterlogging is often addressed **after complaints or disruption**.



Rainfall, incident, GIS and drainage information are **fragmented**.



Limited municipal teams still need to decide **where to act first**.



Repeated failures may hide **upstream drainage causes**.



PROPOSED SOLUTION

Predict → Explain → Prioritize → Optimize → Simulate

CivicPulse combines rainfall, historical incidents, terrain, drainage, recurrence and exposure to predict **24/48/72-hour waterlogging risk**, explain why an area is at risk, estimate exposure, identify probable upstream causes, and generate a **preventive response plan**.



INNOVATION / UNIQUENESS



PRE-COMPLAINT FORECAST

Predict waterlogging risk before complaint volume rises.



DRAINAGE DEPENDENCY GRAPH

Trace risk through drain–road–waterbody relationships.



CREW & BUDGET OPTIMIZATION

Generate a feasible preventive work plan using available crews, time and budget.

TECHNICAL APPROACH



Innovex-1

END-TO-END DATA → AI → DECISION FLOW

DATA SOURCES

- Rainfall + forecast
- Incidents / reports
- Elevation / terrain
- Drainage + roads
- Population + facilities

INGEST & VALIDATE

- CSV / GeoJSON / APIs
- Coordinates + timestamps
- Missing / duplicate checks
- Data quality + provenance

FEATURE ENGINE

- Rainfall lags / anomaly
- Terrain + flow indicators
- Topology + recurrence
- Exposure features

AI & ANALYTICS

- XGBoost / LightGBM
- 24/48/72h risk
- SHAP explanation
- Spatial + graph analysis

DECISION LAYER

- Impact estimation
- Risk + priority
- Top preventive actions
- OR-Tools optimisation

USER / SIMULATION

- Chennai risk map
- Evidence card + action queue
- What-if scenarios
- Officer verification

DATA STORE • POSTGRESQL + POSTGIS • spatial layers, incidents, features, predictions, provenance & audit logs

TECH STACK

- ☐ Next.js + React + TypeScript
- ☐ Tailwind CSS
- ☐ FastAPI + Python
- ☐ PostgreSQL + PostGIS
- ☐ Pandas + NumPy + GeoPandas
- ☐ Leaflet / MapLibre

AI / INTELLIGENCE

- ☐ XGBoost / LightGBM
- ☐ SHAP
- ☐ Spatial-temporal clustering
- ☐ Graph logic
- ☐ Exposure engine
- ☐ OR-Tools optimisation

TRUST / CONTROL

- ☐ Confidence + evidence
- ☐ Public / observed / simulation labels
- ☐ Human approval
- ☐ Forecast refresh
- ☐ Officer override

FEEDBACK LOOP: FIELD OUTCOME → DATA QUALITY → MODEL RECALIBRATION

AI recommendation is decision support — not an automatic municipal order. Every recommendation must remain evidence-backed and field-verifiable.

FEASIBLE NOW**PROTOTYPE SCOPE**

- Chennai-focused MVP; one focused hazard: drainage-related waterlogging.
- 24/48/72h forecast horizons.
- Public/official rainfall + GIS + population/elevation data can seed the prototype.
- CSV/GeoJSON imports handle legacy or incomplete data.

KEY RISKS**Data & operational limits**

- True waterlogging/failure labels may be sparse.
- Drainage condition and ownership can be incomplete.
- Forecast uncertainty changes risk.
- Complaint reporting is not a neutral measure of risk.
- Stale/missing GIS can reduce confidence.

MITIGATION**CONTROLS**

- Baseline + ML comparison; time-based and spatial holdouts.
- Show confidence, evidence and data-quality warnings.
- Separate public / observed / simulation modes.
- Human verification before action; officer override.
- Recalculate as forecasts update.

36-HOUR MVP • RISK MAP • EXPLANATION • EXPOSURE • TOP-5 ACTIONS • CREW OPTIMIZER • WHAT-IF

No 100% accuracy claims. No guaranteed disaster-prevention claims. Data quality determines model reliability.



IMPACT AND BENEFITS



IMPACTS

MUNICIPAL OFFICERS

➤ Earlier risk watchlists and evidence-backed preventive decisions.

FIELD TEAMS

➤ Feasible inspection order based on crews, travel, urgency and budget.

CITIZENS

➤ Earlier preventive action can reduce potential disruption in exposed areas.

PLANNING & BUDGET

➤ Use risk, recurrence and exposure to target limited preventive resources.

BENEFITS

- 1 **EARLIER SIGNAL** — 24/48/72h risk horizons support pre-event inspection.
- 2 **EXPLAINABLE DECISION** — Shows contributing factors, **evidence** and confidence.
- 3 **EXPOSURE-AWARE** — Estimates **population** and **critical-facility** exposure.
- 4 **RESOURCE-AWARE** — Matches **priority** with **available crews, time** and budget.
- 5 **PREPAREDNESS** — Compares **rainfall** and **intervention scenarios** before acting.

PROPOSED PILOT KPIS

Lead time before event • Recall@top-10 risk cells • Risk reduction / budget unit • Exposure reduction • Travel distance saved

RESEARCH BASIS

3 RESEARCH PAPERS

RESEARCH PAPERS

1. **Balamurugan, Choudhary & Raja** — “Prediction of Flooding Due to Heavy Rainfall in India Using Machine Learning Algorithms: Providing Advanced Warning,” **IEEE Systems, Man, and Cybernetics Magazine**, 2022. DOI: [10.1109/MSMC.2022.3183806](https://doi.org/10.1109/MSMC.2022.3183806)
2. **Hassan et al.** — “Machine Learning-Based Rainfall Prediction: Unveiling Insights and Forecasting for Improved Preparedness,” **IEEE Access**, 2023. DOI: [10.1109/ACCESS.2023.3333876](https://doi.org/10.1109/ACCESS.2023.3333876)
3. “**Urban waterlogging prediction and risk analysis based on rainfall time series features**: A case study of Shenzhen,” **Frontiers in Environmental Science**, 2023.
<https://www.frontiersin.org/journals/environmental-science/articles/10.3389/fenvs.2023.1131954/full>

OFFICIAL / DATA SOURCES

OFFICIAL / REAL-WORLD SOURCES

- **India Meteorological Department (IMD)** — rainfall and weather resources.
<https://mausam.imd.gov.in/>
- **Greater Chennai Corporation GIS** — official ward/zone and municipal geospatial layers.
https://gisgcc.chennaicorporation.gov.in/server/rest/services/GCCPublic/GCC_AdminBoundary/MapServer
- **OpenStreetMap** — road, waterway and place context for geospatial analysis.
<https://www.openstreetmap.org/>

DATA NOTE

Public, observed and simulation inputs must be labelled separately. Prototype claims should be replaced by validated measurements before submission.

Reference links verified against the cited source titles.